

Homework 3: Statistics in the Real World

1. This article challenges the long-standing theory that men evolved as hunters and women as gatherers. It argues that this view is not supported by current evidence. Recent studies suggest women may be physiologically better suited for endurance activities, like those required in hunting, due to factors like fat metabolism and muscle composition. Fossil records and ethnographic studies show that women have been involved in hunting throughout history, contradicting the idea of a strict division of labor based on gender. The article discusses how cultural biases have influenced scientific theories and acknowledges the need for more inclusive research that represents both female and male contributions to human evolution.
2. The author's bias is that they argue against the notion of a strict division of labor based on gender. The authors seem to advocate for a view of prehistoric human societies that is more egalitarian and inclusive of women as active participants in hunting. Their approach counters the long-standing bias that underestimates women's physical capabilities and roles in ancient societies.
3. This article has multiple instances of statistical bad behavior. The article challenges the traditional "Man the Hunter" theory, which has been widely accepted in anthropology. By focusing on evidence that counters this theory, the authors may be selectively presenting data that aligns with their argument. If they omit or downplay contrary evidence, it could be considered a form of selection bias. This article may engage in overgeneralization since the article discusses the role of women in hunting throughout history but does not provide specific statistics on the prevalence of female hunters in ancient societies. This article may also engage in reporting bias. While mentioning physiological differences between males and females, the article lacks quantitative data comparing endurance and strength capacities across genders. This article also refers to ethnographic studies of modern hunter-gatherer societies but does not include statistical evidence to support the involvement of women in hunting activities.

Part 2

1. This article discusses the work of behavioral economists Dan Ariely and Francesca Gino, who became well-known for their research on dishonesty and truth-bending. However, they have faced accusations of fabricating data in their studies. The piece delves into Ariely's background, his personal experiences with pain and deception, and how these influenced his academic pursuits. It also touches on the broader field of behavioral economics, its impact on policy and society, and the replication crisis that has called into question many findings in the discipline. The narrative raises questions about the integrity of research and the consequences of potential misconduct in academia. This article also delves into the challenges and controversies surrounding the replication

and validity of a research study on honesty. It highlights the complexities of scientific research and the importance of data integrity.

2)

To address the issues of data fabrication and unethical research practices described in this article, some solutions that I thought of are the implementation of strict oversight mechanisms within academic institutions to monitor research activities and data handling. This could involve regular audits of raw data and methodologies by independent committees. Another solution would maybe be to educate early-career researchers and emphasize the importance of ethical research practices and the long-term consequences of academic dishonesty. Researchers could have ethically searched for positive results by committing to a hypothesis and analysis plan before collecting data to prevent p-hacking and selective reporting of positive results. Researchers can also embrace and publish null results to combat publication bias and the pressure to find significant outcomes, thus providing a more accurate picture of the research landscape.